2946. Matrix Similarity After Cyclic Shifts

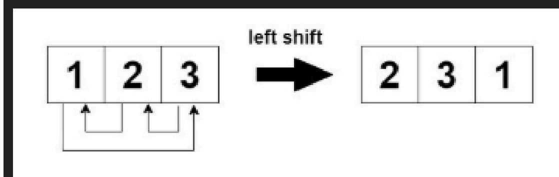
Solved

Easy | Topics | Companies | Hint

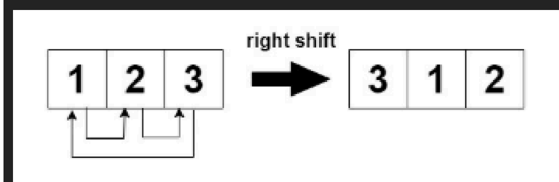
You are given an `m x n` integer matrix `mat` and an integer `k`. The matrix rows are 0-indexed.

The following proccess happens `k` times:

- **Even-indexed** rows (0, 2, 4, ...) are cyclically shifted to the left.



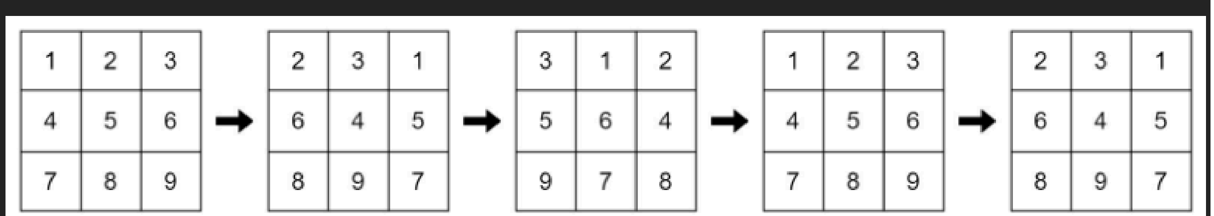
- **Odd-indexed** rows (1, 3, 5, ...) are cyclically shifted to the right.



Return `true` if the final modified matrix after `k` steps is identical to the original matrix, and `false` otherwise.

Example 1:

> **Input:** `mat = [[1,2,3],[4,5,6],[7,8,9]], k = 4`
>
> **Output:** `false`
>
> **Explanation:**
>
> In each step left shift is applied to rows 0 and 2 (even indices), and right shift to row 1 (odd index).



Example 2:

> **Input:** `mat = [[1,2,1,2],[5,5,5,5],[6,3,6,3]], k = 2`
>
> **Output:** `true`
>
> **Explanation:**
>
> $$\begin{bmatrix}1 & 2 & 1 & 2\\ 5 & 5 & 5 & 5\\ 6 & 3 & 6 & 3\end{bmatrix} \rightarrow \begin{bmatrix}2 & 1 & 2 & 1\\ 5 & 5 & 5 & 5\\ 3 & 6 & 3 & 6\end{bmatrix} \rightarrow \begin{bmatrix}1 & 2 & 1 & 2\\ 5 & 5 & 5 & 5\\ 6 & 3 & 6 & 3\end{bmatrix}$$

Example 3:

> **Input:** `mat = [[2,2],[2,2]], k = 3`
>
> **Output:** `true`
>
> **Explanation:**
>
> As all the values are equal in the matrix, even after performing cyclic shifts the matrix will remain the same.

Constraints:

- `1 <= mat.length <= 25`
- `1 <= mat[i].length <= 25`
- `1 <= mat[i][j] <= 25`
- `1 <= k <= 50`

Python:

```
class Solution:
    def areSimilar(self, mat, k):
        m, n = len(mat), len(mat[0])

        k %= n  # (reduce k<n)
```

```
        for i in range(m):
            for j in range(n):
                if i % 2 == 0:
                    # even row , left shift
                    if mat[i][j] != mat[i][(j + k) % n]:
                        return False
                else:
                    # odd row , right shift
                    if mat[i][j] != mat[i][(j - k) % n]:
                        return False

        return True
```

JavaScript:

```
var areSimilar = function(mat, k) {
    let m = mat.length, n = mat[0].length;
    k %= n;

    for (let i = 0; i < m; i++) {
        for (let j = 0; j < n; j++) {
            let newCol;
            if (i % 2 === 0)
                newCol = (j + k) % n;
            else
                newCol = (j - k % n + n) % n;

            if (mat[i][j] !== mat[i][newCol])
                return false;
        }
    }
    return true;
};
```

Java:

```
class Solution {
    public boolean areSimilar(int[][] mat, int k) {
        int m = mat.length;
        int n = mat[0].length;

        k %= n; //(reduce k<n)

        for (int i = 0; i < m; i++) {
            for (int j = 0; j < n; j++) {
                if (i % 2 == 0) {
                    // even row , left shift
```

```
                if (mat[i][j] != mat[i][(j + k) % n]) {
                    return false;
                }
            } else {
                // odd row , right shift
                if (mat[i][j] != mat[i][(j - k + n) % n]) {
                    return false;
                }
            }
        }
    }

    return true;
  }
}
```